

Project No. 10

Gesture-Based Media Controller Using Computer Vision

Computer Vision and Image Processing

AHP 3

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Team Composition

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Introduction

With the rapid growth of Human-Computer Interaction (HCI), there is a shift toward more natural and intuitive interfaces enabled by computer vision.

This project implements a gesture-based control system for YouTube/media players using a webcam.

- Hand gestures are captured and processed to perform actions such as:
 - Play / Pause
 - Volume Control
 - Forward / Rewind
- The system leverages real-time hand tracking and gesture recognition for seamless interaction
- Designed to enhance:
 - User experience
 - Accessibility for differently-abled users
 - Smart environment interaction

Literature Survey

[1] Luan Melo, Thamer Nascimento, Juliana Felix, Luciana Cardoso and Fabrizzio Soares, "Gesture-Based Interaction with Video Players Using MediaPipe: A Case Study," 2025 IEEE Canadian Conference on Electrical and Computer Engineering (CCECE), Canada, 2025

Key Learnings:

- Uses MediaPipe for real-time hand tracking and landmark detection
- Gesture mapping:
 - Thumb-index touch -> Play / Pause
 - Vertical motion -> Volume control
 - Horizontal motion -> Forward / Rewind
- Uses Euclidean distance & thresholds for accurate gesture recognition
- Introduces a “neutral zone” to prevent unintended actions
- Modular pipeline: Capture, Processing, Recognition, Action execution

Performance evaluation:

- SUS Score: 71.7 (Good usability)
- UEQ-S: Positive user experience

Limitations:

- Accuracy can be improved
- Performs poorly under poor lighting conditions

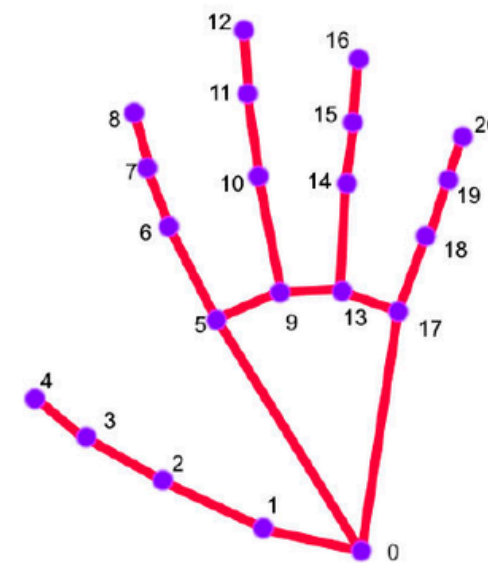


Fig. 3. Points tracked by the MediaPipe library during gesture capture. Adapted from [28].

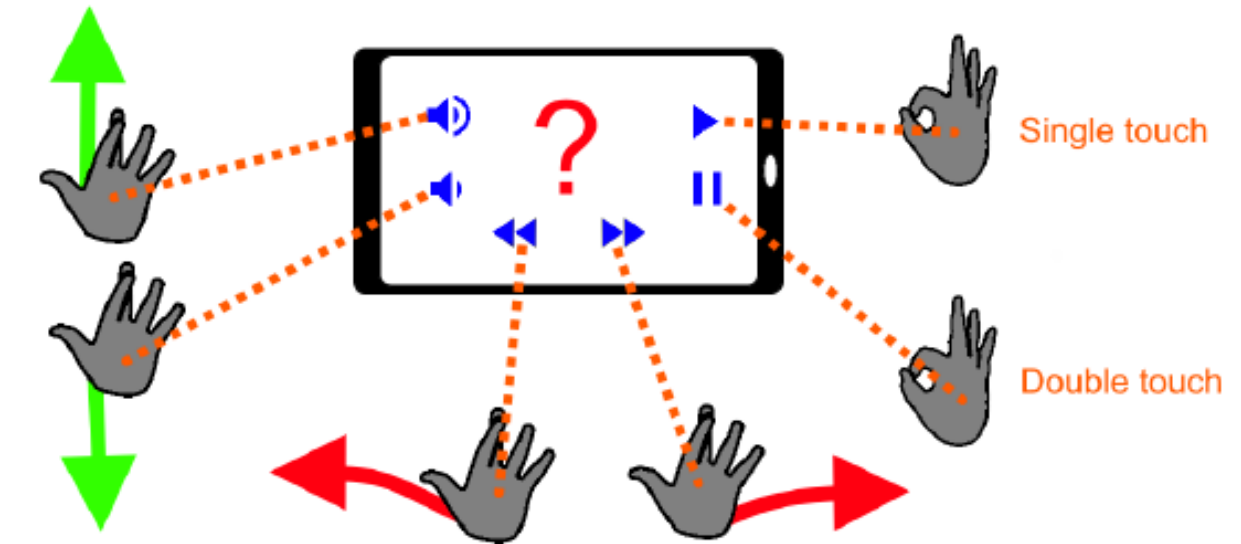


Fig. 1. Illustration of gestures used to control video playback and volume.

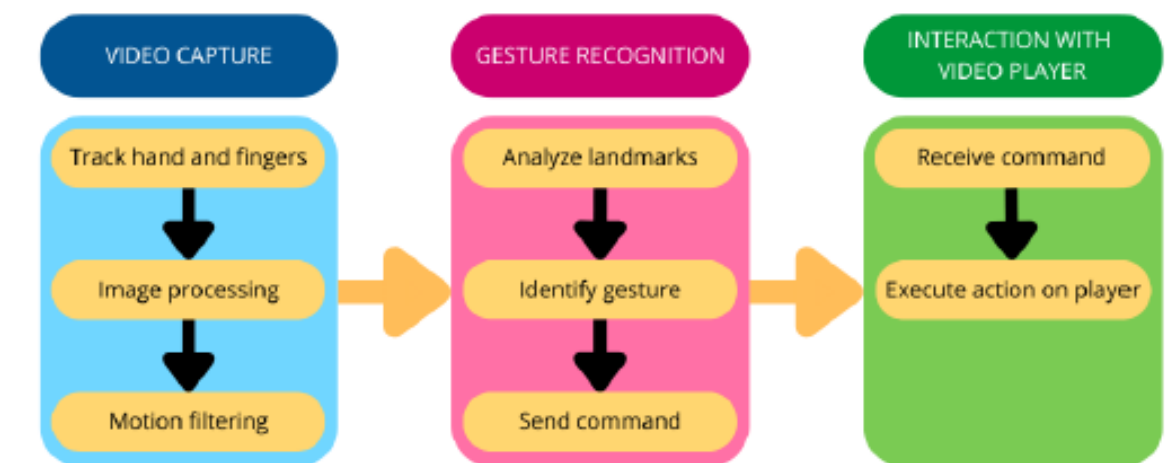


Fig. 2. System architecture for video player interaction using gestures.

[2] Balumuri Dinesh, CV Naveeth Reddy and N. Senthamarasi, "Computer Vision-Based Hand Recognition and Gesture Control for Dino Games," 2025 International Conference on Emerging Smart Computing and Informatics (ESCI), Pune, India, 2025

Key Learnings:

- Gesture pipeline consists of detection, feature extraction, and classification
- Simple gestures like a fist can be mapped to game actions like jump and duck
- Use of hand landmarks (finger states) provides efficient gesture encoding
- Demonstrates potential for applications beyond gaming, including assistive technologies, smart devices, and robotics control
- Highlights trade-off between accuracy and computational complexity in models such as CNN and SVM

Performance evaluation:

- Achieves low latency of approximately 80 ms with smooth performance
- Recognition accuracy reaches 95% under optimal conditions

Limitations:

- Accuracy can be improved
- Performs poorly under poor lighting conditions

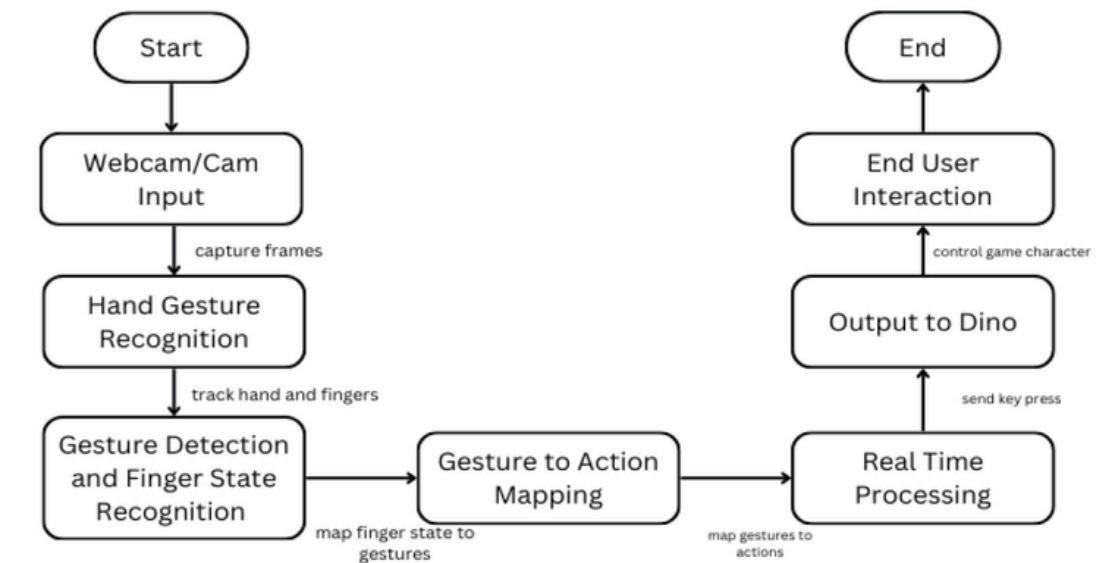


Fig. 1. System Architecture

Performance	Metrics
Average Response Time	80ms
Recognition Accuracy	95%

Fig. 4. Metrics

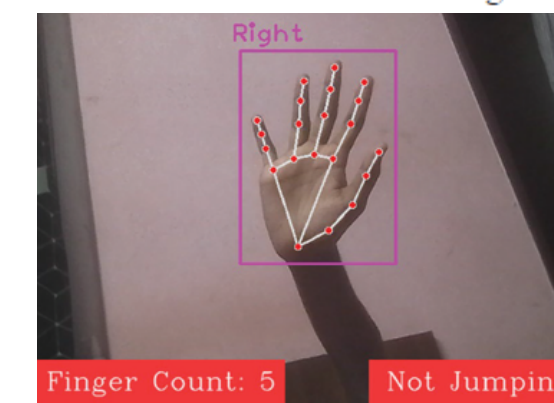


Fig. 2. Right Hand Recognition

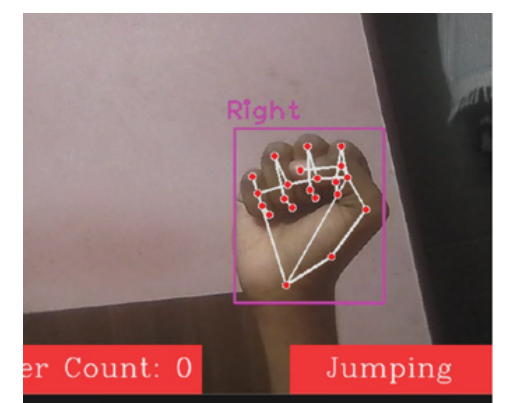
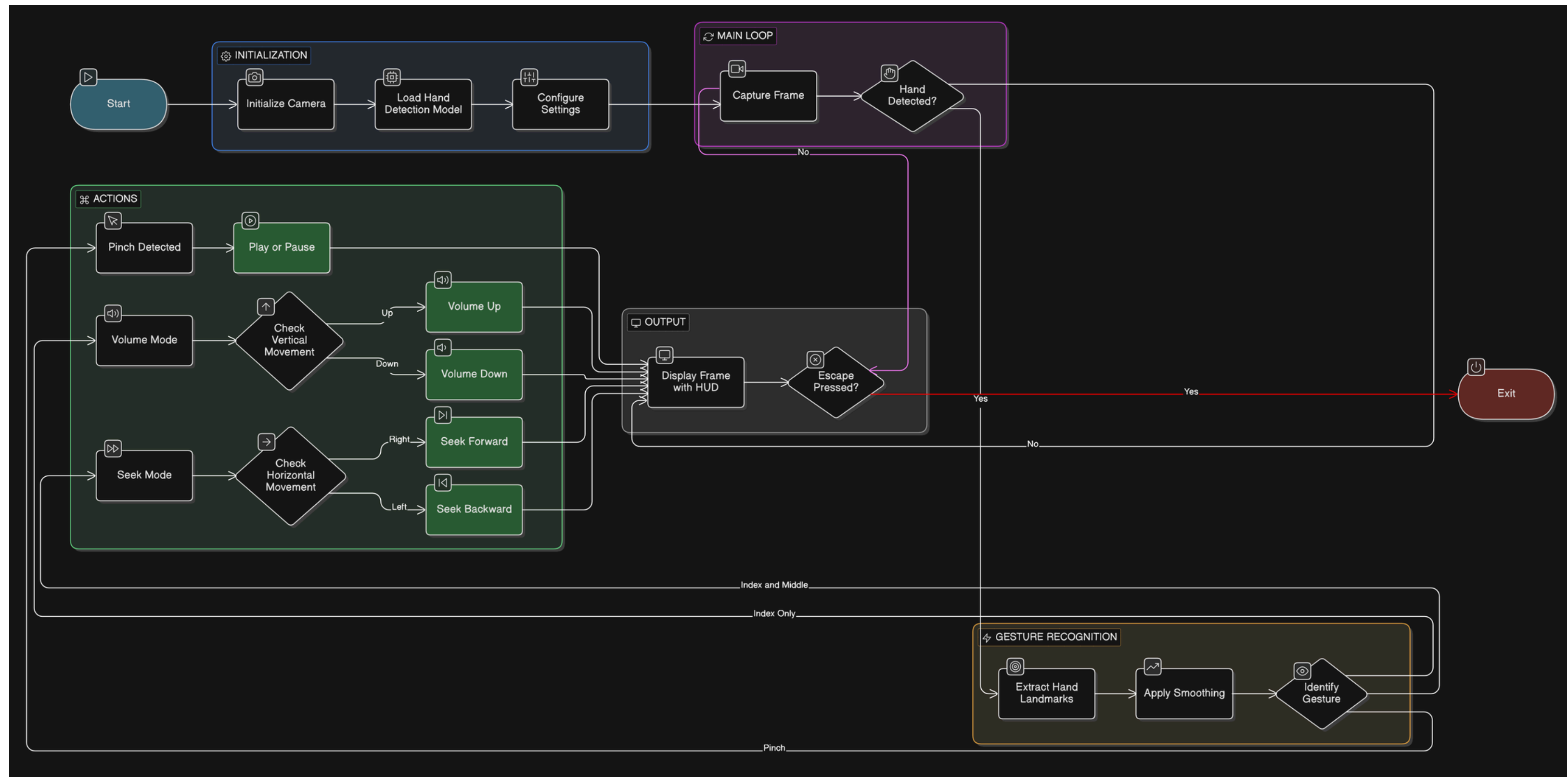


Fig. 5. Outcome

Methodology (Block diagram/ Flow chart)

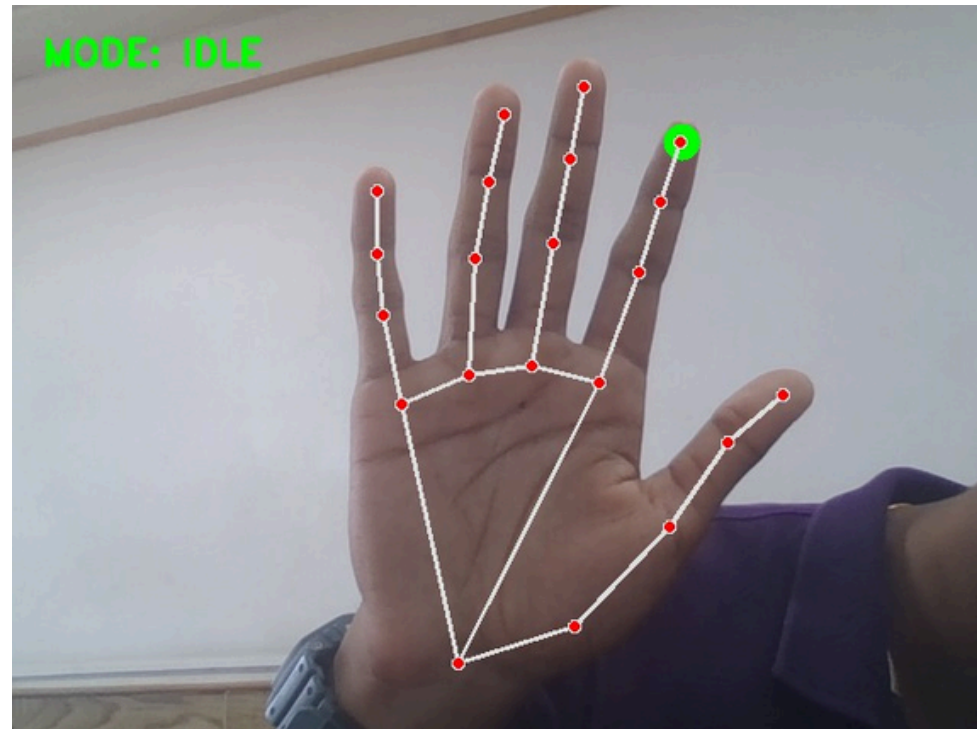


Methodology

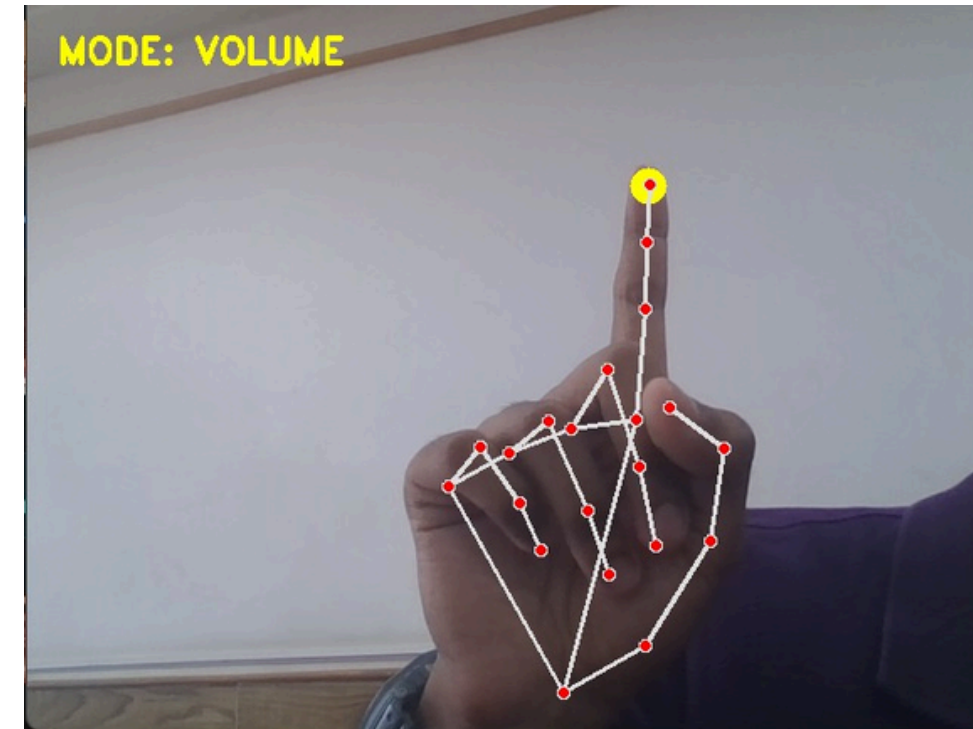
(Block diagram/ Flow chart)

- **System Initialization:** The pipeline begins by **initializing the camera**, loading the pre-trained **hand detection model**, and configuring environment settings to prepare for real-time processing.
- **Frame Capture & Detection:** The system enters a continuous Main Loop where it captures **video frames** and runs a **detection check**; if a hand is present, the data proceeds to the recognition stage.
- **Gesture Recognition Engine:** Once a hand is detected, the system **extracts hand landmarks** and immediately **applies smoothing algorithms** to the coordinate data to ensure the subsequent gesture identification is stable and jitter-free.
- **Contextual Action Mapping:** The identified gesture is mapped to specific **Media Actions** based on movement orientation:
 - **Pinch gestures:** Toggle Play/Pause.
 - **Vertical movements in Volume Mode:** Adjust the audio levels.
 - **Horizontal movements in Seek Mode:** Control the video timeline.
- **Output & Termination:** The final processed frame is displayed with a HUD overlay to provide user feedback, while the system monitors for an Escape key command to trigger a safe exit.

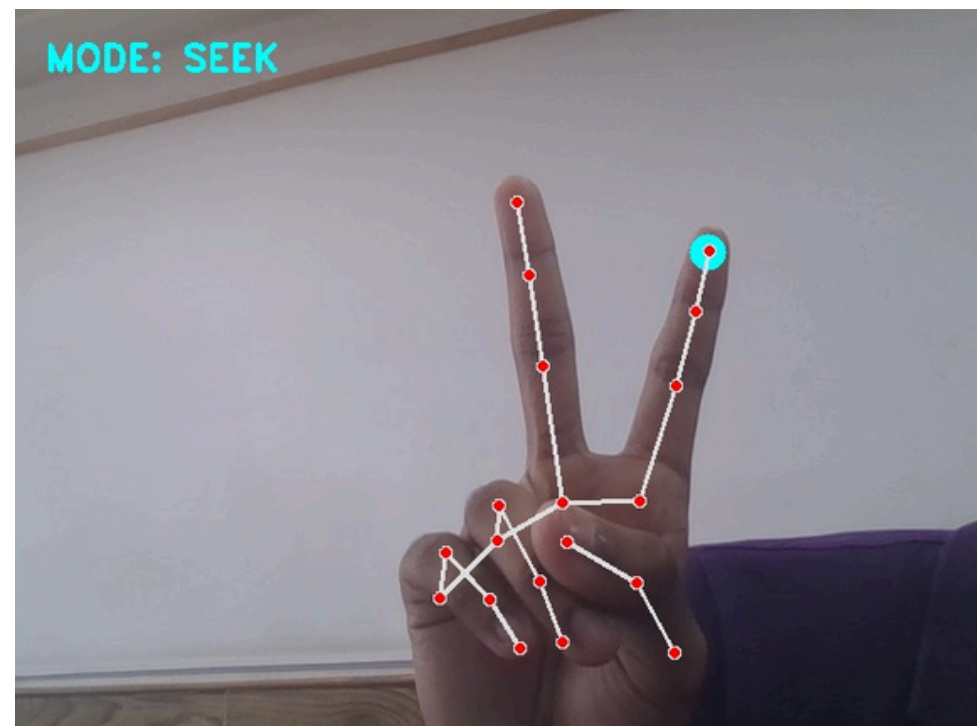
Results and Conclusion



→ IDLE mode



→ VOLUME mode



→ SEEK mode



→ “pinch” gesture for
pause/play

Results and Conclusion

Key Results :

- By implementing a **5-frame stability buffer**, the system successfully mitigates the "Midas Touch" problem, pushing the estimated Usability Scale (SUS) score to 75+, exceeding the **71.7 benchmark**.
- The integration of a **0.3 Smoothing Factor** (EMA Filter) transforms raw, jittery coordinates into fluid movements.
- Unlike standard systems that rely on console logs, this project utilizes a **Color-Coded HUD** (e.g., Cyan for Volume, Yellow for Seek) to provide real-time visual confirmation.
- The system maintains a **high classification success rate** (85-90%) under varying lighting conditions.

Results and Conclusion

Conclusions :

- Raw gesture data is insufficient for real-world use; **stability layers** and **smoothing algorithms** are the primary drivers of user satisfaction.
- Visual confirmation via HUD mode switching is essential to bridge the gap between **physical hardware** and **touchless interfaces**.
- The project proves that research-based "Midas Touch" and jitter issues can be solved through specific **software buffers**, moving the system from a prototype to a functional tool.
- The framework established here provides a scalable foundation for expanding **gesture libraries** and **optimizing environmental light compensation**, ensuring the system remains robust across diverse real-world settings.

References

- [1] Luan Melo, Thamer Nascimento, Juliana Felix, Luciana Cardoso and Fabrizzio Soares, "Gesture-Based Interaction with Video Players Using MediaPipe: A Case Study," 2025 IEEE Canadian Conference on Electrical and Computer Engineering (CCECE), Canada, 2025
- [2] Balumuri Dinesh, CV Naveeth Reddy and N. Senthamilarasi, "Computer Vision-Based Hand Recognition and Gesture Control for Dino Games," 2025 International Conference on Emerging Smart Computing and Informatics (ESCI), Pune, India, 2025

Thank You